

Review Article

Bacterial-based drug delivery systems: A new way to combat infectious disease

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ABSTRACT

Recent advances in targeted drug delivery system (DDS) have generated high expectations for the treatment of various diseases. The main advantages of DDS are precise and efficient drug delivery, increased concentration and effectiveness of drugs at the site of action, minimal systemic distribution of harmful drugs, and reduced side effects. The choice of DDS often depends on the specific disease being treated, with common options including liposomes, nanoparticles, microspheres, and various biomaterials such as cell lines and microbial components.

DDS is most prominent in cancer therapy, where challenges such as limited access to tumor tissues and drug-inactivating environments complicate treatment. However, in recent decades, the emergence of antibiotic resistance and the therapeutic difficulties associated with chronic and intracellular bacterial infections have made infectious diseases a key focus for DDS development. Initial DDS approaches for bacterial infections were based on nano-derivatives. During more advanced stages of research, even the bacteria themselves became vehicles for DDS focused on microbial infections. All aspects of the bacteria- cell lysates, envelopes, derived vesicles, spores, and so forth-have been evaluated as DDS. This form of DDS does not only promote the immediate treatment of an infection but also expedites recovery times through immunological manipulation. Despite these advancements, there remains a lack of cohesive data regarding bacterial DDS in the treatment of infectious diseases. Therefore, this review aims to provide an overview of the various types of bacterial DDS, their applications, advancements, and the challenges they face.

1. Introduction

Infectious diseases are the second leading cause of death worldwide, following heart disease. Treating microbial infections presents many challenges, including the spread of antibiotic resistance, genetic mutations and adaptations of pathogens to their hosts, the existence of asymptomatic carriers, zoonotic disease cycles, and systemic involvement in individuals with underlying conditions. Therefore, the World Health Organization (WHO) has long considered in-depth research on new approaches to fighting infectious diseases a priority [1]. Many studies have been performed with the aim of finding antibiotic alternatives. Some of the most promising candidates include essential oils, antimicrobial nanoparticles, and bacteriophages [2]. Other studies involved patient support strategies and infection control measures in medical centers and communities. Additionally, the development of various microbial vaccines has been explored to help manage infectious diseases [3]. Table 1.

Recent strategies focused on using drug delivery systems (DDSs) to target infections. These approaches involve the use of nanotechnology, liposomes, microparticles, and biological drug carriers to address microbial infections. DDSs offer several advantages, including targeted drug delivery, controlled drug release, and immune system modulation.

When used as drug carriers, the delivery of small drugs like nanomedicines or gene therapy agents can be stimulated by the immune system through microbial cells or their derivatives. Specifically, bacterial DDSs have shown therapeutic capacity in the treatment of infections [4]. But many studies focused on cancer treatment by bacterial DDSs, the treatment of infectious diseases has received less attention. This review aims to emphasize the potential of bacterial DDSs as innovative approaches to treating infections. Therefore, we first described DDSs, followed by an exploration of bacterial DDS types, their applications, and the challenges they face in treating infectious diseases.

Table 1

Examples of different routes for bacterial DDSs.

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Delivery method	Description	Examples and notes
Intravenous Injection (i. v.)	Allows bacteria to circulate throughout the body, reaching various tumor sites via the bloodstream.	Used for delivering engineered <i>Salmonella typhimurium</i> (SAM-FC) in studies.
Intra-tumoral Injection (i. t.)	Maximizes bacterial concentration at the target site, enhancing therapeutic effectiveness.	Direct injection into tumors to minimize systemic exposure.
Oral Delivery	Challenges due to acidic gastrointestinal conditions; advancements in encapsulation show potential.	Research is ongoing for effective oral administration of engineered bacteria.
Combination Therapies	Combines oncolytic bacteria with conventional chemotherapy or immunotherapy to boost efficacy.	Example: <i>monocytogenes</i> in melanoma clinical trials led to immune response and tumor shrinkage.
Bacterial Effector Proteins	Utilizes proteins like HopQ from <i>Pseudomonas syringae</i> to target cancer cells directly.	HopQ binds to vimentin in melanoma cells, reducing mobility and metastasis through selective autophagy.
Heparin Production	Engineered <i>E. coli</i> K5 used for producing heparin from N-trifluoro-acetyl- glucosamine.	Heparin is crucial for anticoagulation, administered subcutaneously or intravenously. Production requires purification for use.

2. Drug delivery systems

2.1. Definition and history

DDSs represent a critical integration of pharmacology, engineering, and biotechnology, designed to ensure medications reach their intended targets effectively and safely. These systems are developed to enhance the therapeutic effects of drugs while minimizing side effects, ultimately improving patient outcomes. The systematic extraction of active compounds dates back to ancient Egyptian and Greek civilizations, marking the beginning of drug formulation techniques [5]. The study of early medicinal practices across various cultures reveals a wealth of knowledge that has shaped contemporary medicine. The reliance on natural substances such as plants, minerals, and animal products demonstrate humanity’s deep connection to the environment and the resources it offers. This relationship continues to influence modern medical practices and philosophies, as many plants still provide valuable medicinal benefits [6].

In the 19th century, advancements in chemistry allowed for the isolation of active pharmaceutical ingredients (APIs), leading to the development of various dosage forms, including syrups and pills. These forms improved the consistency and efficacy of treatments [7]. The introduction of controlled-release formulations in the mid-20th century marked a significant advancement in drug delivery, enabling constant drug release over extended periods. Although initial release rates were limited, they improved as research progressed. Injectable formulations, driven by the development of vaccines and antibiotics, enhanced the speed and efficiency of drug administration. Other examples are transdermal patches, liposomes, and biodegradable polymers, revolutionized the discipline of drug delivery. These technologies facilitated sustained and targeted drug release, thereby improving therapeutic efficacy and compliance. Continuation research in DDSs has led to the creation of systems from cells and cell components for targeting specific diseases via targeted routes, such as the oral microsphere formulation of M2 macrophage-mimetic nanomotors for the targeted treatment of ulcerative colitis, local controlled-release systems for therapeutic

proteins in skeletal muscle injuries, and controlled-release mitochondrial protonophores (CRMP) to treat dyslipidemia and hepatic steatosis in dysmetabolic nonhuman primates [8]. Fig. 1 provides an overview of the history of drug delivery developments since 1950 [9]. In recent years, there has been increasing interest in the use of peptides and microbial DDSs, particularly bacteriophages (Fig. 1).

2.2. Advantages and applications

DDSs offer several advantages that improve the effectiveness and safety of the treatment. One primary benefit is targeted delivery, which directs drugs precisely to the site of action, minimizing side effects and improving therapeutic outcomes. Therefore, a lower dose of drug is needed and it is another advantage of DDSs, the reduction of drug waste. DDS also enhance the bioavailability of drugs, particularly those with poor solubility, thus increasing their absorption and effectiveness. An example of this is the use of water-soluble metalated covalent organic nanobelts to improve bioavailability for protein transport [10]. More uniform drug effects, more safety and less plasma level fluctuations are other DDS advantages. These benefits make DDS a critical component in modern pharmacotherapy, contributing to more effective and safer treatment options. Targeted drug delivery is useful in several fields, including cancer treatment, inflammation, infectious diseases, and reproductive health. Various current targeted delivery methods have shown promising results, including cryo-shocked cancer cells for targeted delivery and vaccination [11], enzyme-powered Janus platelet cell robots for active drug delivery, autonomous metal-organic framework nanorobots for mitochondria-targeted cancer therapy [12], genetically engineered cell membrane-coated nanoparticles for targeted delivery of dexamethasone to inflamed lungs, and self-adaptive micro/ nanorobots for gastrointestinal inflammation therapy [13]. These methods have demonstrated efficacy across various applications.

2.3. Different types of drug delivery systems

There are several types of DDSs, each related to specific applications and technologies (Fig. 2). Conventional DDS, such as tablets, creams, and solutions, serve to increase the dosage of active pharmaceutical ingredients to maintain effective levels of the drug [14]. Controlled-release DDSs aim to keep drug plasma levels stable by administering a specific dose at scheduled intervals, reducing both the amount of medication required and the frequency of administration. This approach improves patient adherence and minimizes the risk of toxicity and side effects, thus enhancing overall drug effectiveness. Controlled-release DDSs can be categorized based on the mechanisms of drug release, including dissolution-controlled systems, diffusion-controlled systems, and water penetration-controlled systems, which can further be classified into osmotic pressure-controlled and swelling-controlled types. Other categories include chemically controlled systems and those utilizing nanoparticles for drug delivery [15].

Biomaterials play a key role in controlled-release DDSs by influencing the pharmacokinetics of drugs. These materials are specifically designed to interact with biological systems for therapeutic or diagnostic purposes. Selecting appropriate polymers or biomaterials is essential for creating DDSs with specific physicochemical properties and controlled drug release patterns. Efficient biomaterials in DDS systems include temperature-responsive materials for gastrointestinal applications and electronic proton-trapping ion pumps for selective drug delivery [16]. Stimuli-responsive DDSs have advanced significantly due to the development of biomaterials that react to changes in the external environment. These biomaterials can modify their chemical, physical, and biological characteristics in response to various environmental stimulators. pH-responsive biomaterials are among the most commonly used for controlled drug delivery, as they alter their polymeric structure in response to pH

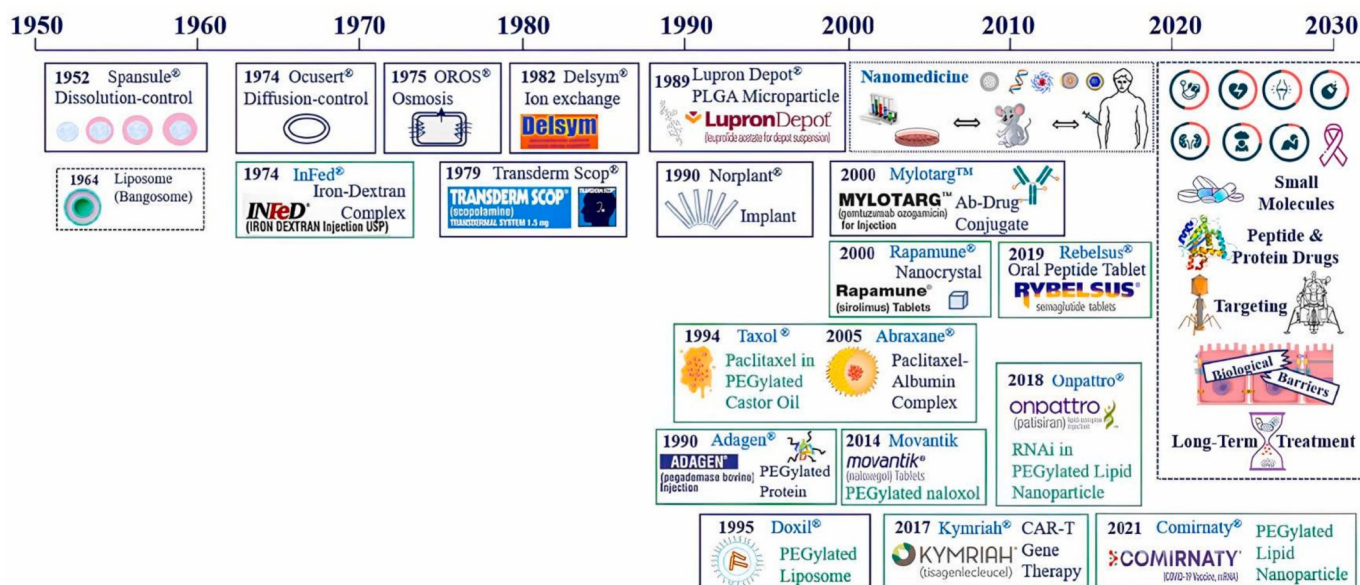


Fig. 1.

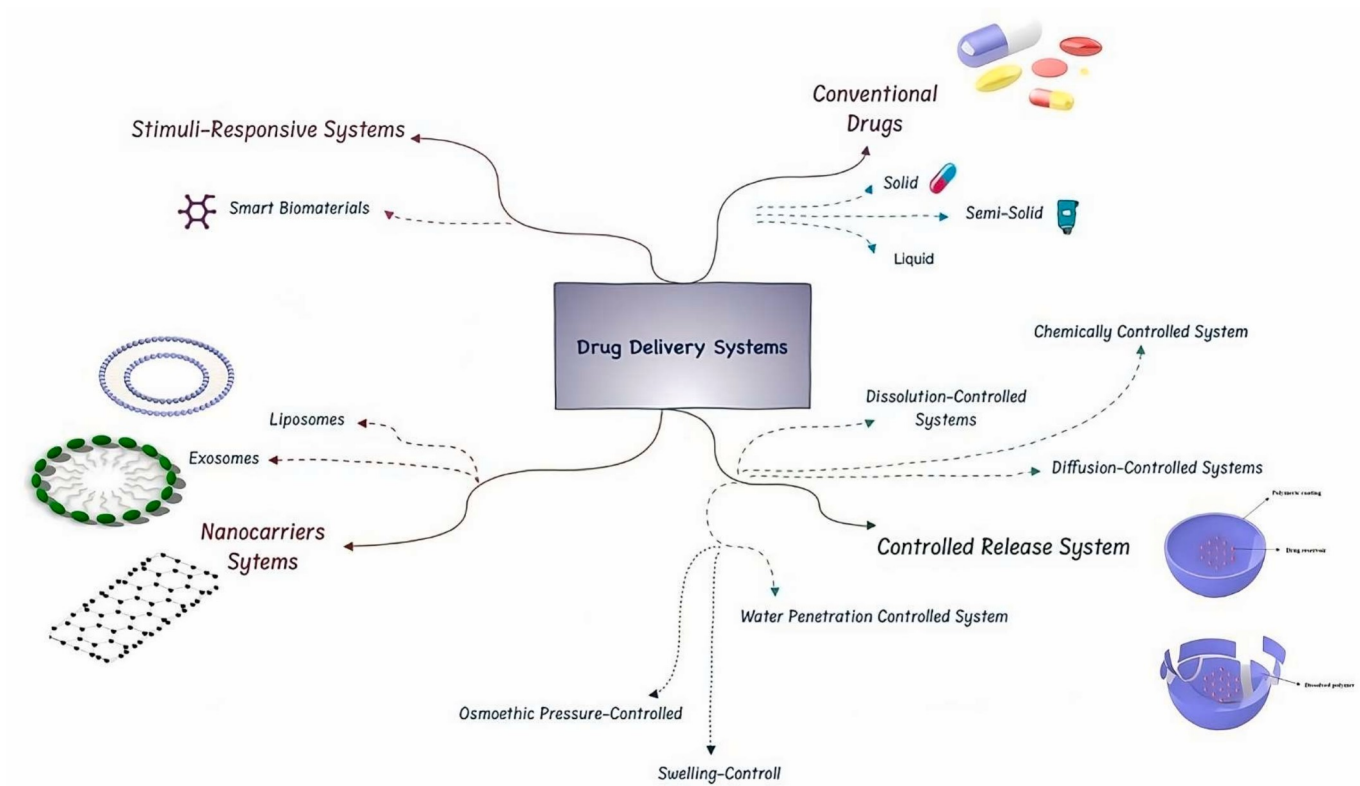


Fig. 2.

changes, initiating drug release. Redox-responsive and enzyme-responsive biomaterials are other examples of smart biomaterials used in stimuli-responsive DDS [17]. The different biomaterials used in the manufacture of various DDSs as well as drug release activators [18] are shown in Fig. 3.

Nanocarriers are submicron particles with large surface areas that allow for higher drug loading per unit volume. Various types of nanoparticles in different forms such as polymers, dendrimers, liposomes, nanogels, carbon tubes, and drug-conjugated components can be used as DDS. They improve drug bioavailability by ensuring drugs remain in circulation longer than the drug alone. Nanocarriers also protect drugs from premature degradation in the body and help deliver therapeutic responses at lower doses, reducing potential side effects. Their surface chemistry can be modified for various drugs and targeting agents, making them versatile in drug delivery. Nanocarriers enable sustained and targeted drug delivery, and they can be directed to specific cell types or regions within cells, such as the nucleus for gene therapy. Examples of effective nanocarrier methods include liposomes for tumor repolarization, engineered exosome-like vesicles for targeted cancer therapy, and stem cell-derived exosomes for treating breast cancer stem cells with VCR sulfate [16].

2.4. Drug-Loading strategies of bacterial carriers

There are two main approaches for loading drugs onto bacterial carriers: surface loading and internal loading (Fig. 4). Each method has distinct mechanisms and advantages. Surface loading involves techniques such as chemical bonding, linker grafting, or physical adsorption. Chemical bonding uses surface groups like amine, sulfhydryl, or hydroxyl to form covalent bonds with drugs, ensuring stable attachment.

For example, in a study using N-acetylmuramic acid, a component of the bacterial cell wall, to chemically modify and attach magnetic Fe_3O_4 nanoparticles onto engineered *Escherichia coli*, which were then used for tumor treatment through fenton-like reactions and localized production of H_2O_2 [19]. Linker grafting employs noncovalent interactions like biotin-streptavidin or antigen-antibody interactions for stable and targeted delivery. In another study, using *Listeria* to accumulate high levels of ^{188}Re in the tumor microenvironment, reducing pancreatic cancer via local radioactivity [20]. Physical adsorption relies on electrostatic interactions, where positively charged drugs bind to the negatively charged bacterial surface. For example, an oral DNA vaccine system using positively charged β -cyclodextrin-

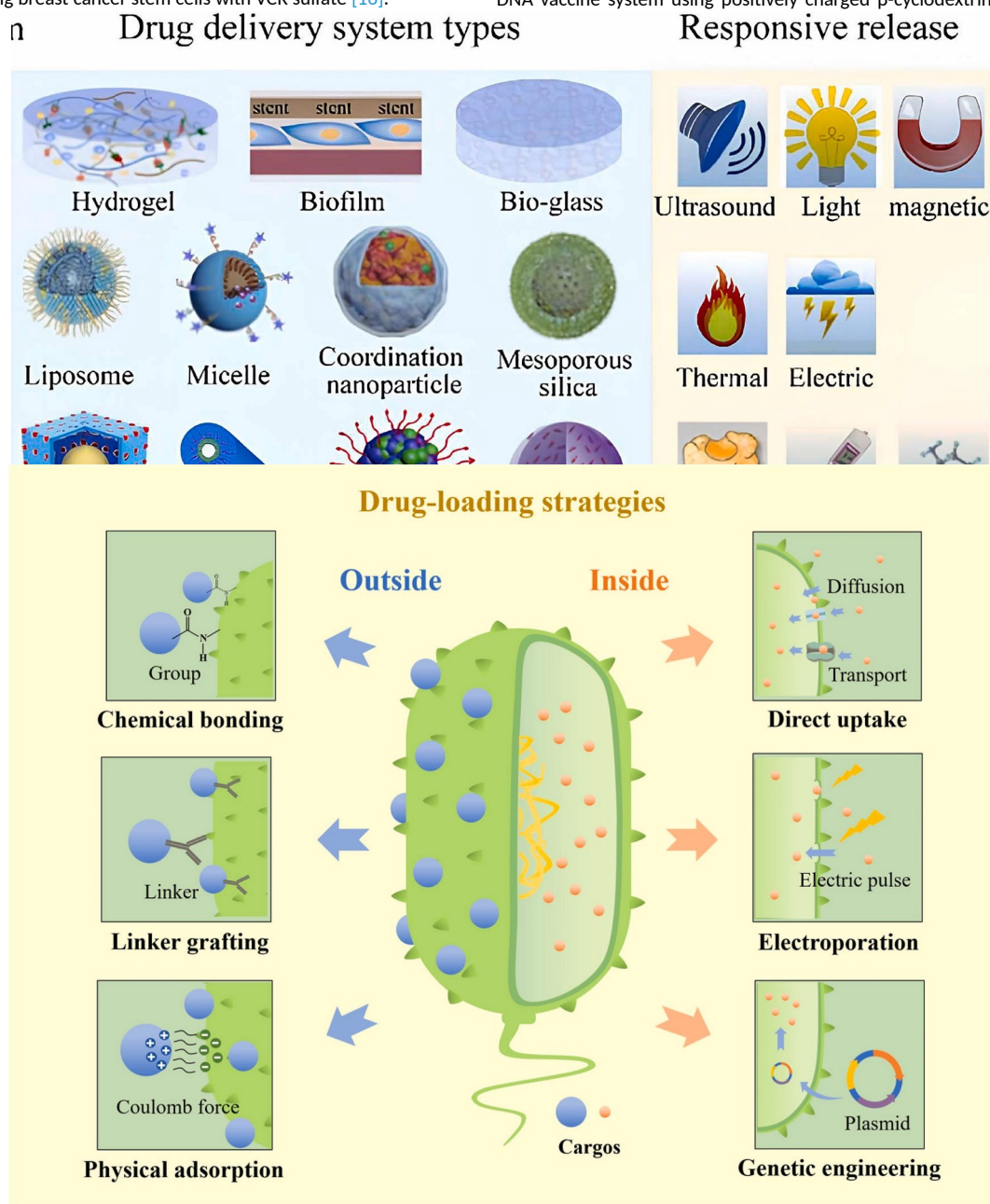


Fig. 4.

PEI600/DNA nanoparticles, which adhered to *Salmonella*, made enhancing bacterial resistance to acidic conditions in the gastrointestinal tract and enabling effective bloodstream entry [21]. Internal loading, achieved either by direct insertion or genetic engineering, differs from surface loading in the way drugs are released. Direct loading delivers drugs in a one-time release, typically through bacterial lysis, while genetically engineered bacteria can continuously synthesize anticancer agents [22].

3. Bacteria and derivatives as drug carriers

Various studies have evaluated bacterial cells and their derivatives as biological DDSs. These can include whole microbial cells, cell lysates, vesicles and membrane derivatives, spores, toxins, and bacterial enzymes. Each type will be discussed separately in this section.

3.1. Whole bacterial cell

Over a century ago, Paul Ehrlich introduced the concept of a “magic bullet”—a treatment designed to precisely target and eliminate disease-causing organisms. Therefore, many scientists have focused on using bacteria as DDSs, especially for combating infectious diseases. A deep understanding of the unique characteristics of bacteria is essential for their therapeutic applications. Using bacteria as delivery vehicles requires exploring the various therapeutic components they can transport. Potential components include proteins, peptides, enzymes, nanofibers, antibodies, and other biologically active molecules, selected based on their therapeutic potential and compatibility with the bacterial vector [23].

Furthermore, engineered bacteria can act as DDSs with specific drug-producing abilities. For example, engineered *E. coli* DH5a-lux/BG carried an enzyme β -glucuronidase to the tumor site to activate a prodrug that attacks cancer cells. The results showed that the engineered bacteria successfully targeted tumor sites while minimizing damage to surrounding healthy tissue [24]. In another clinical experiment, *Lactococcus lactis* was genetically engineered to carry and secrete a therapeutic protein called hTFF1 (human trefoil factor 1) directly onto damaged mucosal tissue. The protein acted locally to protect, heal, and reduce the severity of mucositis. This resulted in a safe and effective treatment, as no bacteria were detected in the bloodstream, demonstrating no systemic risk, and no bacterial accumulation was observed, indicating safe clearance of the engineered bacteria [25]. Fig. 5 shows the pathway of drug synthesis and release in an engineered bacterium as a DDS in the host body (Fig. 5) [26].

Another application involves bacteria's ability to carry therapeutic proteins, such as antibodies, directly to targets. This, combined with their generally recognized as safe (GRAS) status, has paved the way for advanced therapeutic strategies. For example, *Lactobacillus reuteri* was genetically engineered to carry and produce the chemokine CXCL12, known to have beneficial effects on wound healing. These bacterial vectors promoted faster healing by stimulating immune cells and other factors that aid in tissue repair in mice. Following the success of this animal study, a clinical trial was performed, and the results showed that the novel platform used to deliver CXCL12- α could be effective in clinical settings [27].

Another group of bacterial vectors are selected for their natural presence in the human microbiome and their production of natural antimicrobial and anti-inflammatory compounds. For example, *E. coli* Nissle 1917 (EcN) is valued for its anti-inflammatory effects and its ability to induce gut immune defenses, making it a subject of many research studies [28]. Researchers have engineered EcN to act as a delivery vehicle for antimicrobial peptides (AMPs) specifically targeting vancomycin-resistant *Enterococcus* (VRE). The engineered EcN strain has been modified to produce and secrete three specific bacteriocins – Enterocin A, Enterocin B, and Hircin JM79- which have shown potent activity against VRE [29].

3.2. Bacterial lysates

Bacterial lysate refers to the contents released when engineered bacteria are intentionally lysed, or burst open. This process releases cellular components such as proteins, nucleic acids, and metabolites. Due to their high efficiency and well-characterized molecular components, bacterial lysates are used for various purposes, including drug delivery and immune response studies [30]. Several methods exist for producing bacterial lysate on a large scale. One common method is ultrasonic disruption, which uses high-frequency sound waves to break open bacterial cells. Another method is biochemical disruption, known as the LoFT method, which relies on lysozyme and a freeze-thaw cycle to achieve cell lysis. A third method involves the freeze-drying and rehydration of autolysis strains. This process involves transforming *E. coli* cells with the R gene, which encodes the enzyme endolysin. These cells are freeze-dried along with transcription-translation reaction components, and upon rehydration, the cells lyse, initiating cell-free expression. Finally, the freeze-thaw method for producing cell debris-free lysate involves growing and harvesting an autolysis strain of *E. coli* or other gram-negative bacteria, freezing the cells at -80

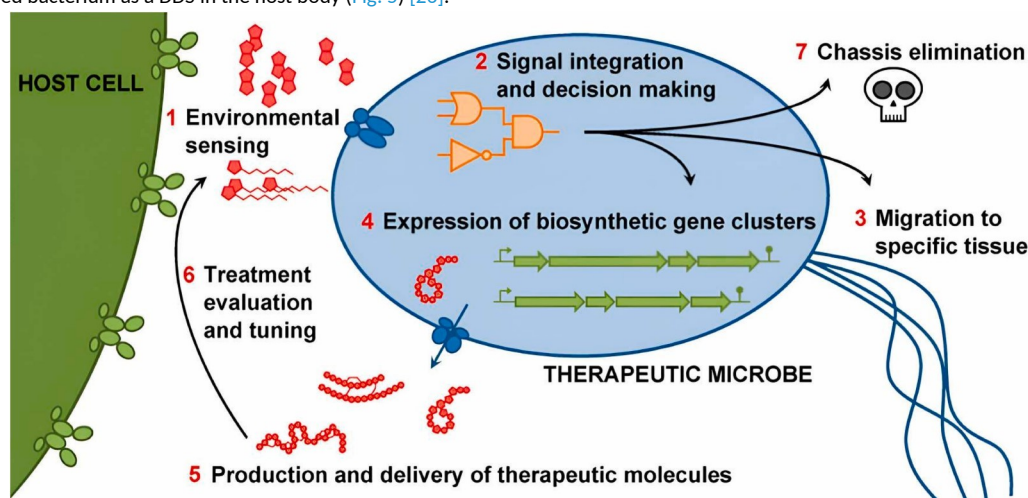


Fig. 5.

°C, thawing them, and then centrifuging to remove cell debris [31].

Bacterial lysates have a wide range of applications across various fields, particularly in medicine and research. They are commonly used as immunostimulants for vaccine development and therapeutic purposes. Additionally, they play a role in DDSs by improving the delivery of therapeutic agents or triggering immune responses [32]. A new example is polyvalent mechanical bacterial lysates (PMBL) DDSs, especially for clinical uses. In a clinical trial, PMBLs of *Staphylococcus aureus* colonize the nasal cavity and help reduce the symptoms of seasonal allergic rhinitis (SAR) by promoting local inflammation and toxin production, leading to a reduction in systemic allergic cytokines [33]. In another clinical trial, the effectiveness of PMBLs in reducing chronic obstructive pulmonary disease (COPD) symptoms over a 12-month period was tested. PMBLs consisted of bacterial strains commonly associated with respiratory infections, including *S. aureus*, *Streptococcus viridans*, and *Klebsiella pneumoniae*. These lysates stimulate the immune system by improving antigen presentation and increasing NK cells, which boost immune responses [34].

Another example is a clinical trial of OM-85, a commercial bacterial lysate, tested against several bacteria that cause respiratory tract infections (RTIs). Since lysates contain various fragments of bacterial cells, such as lipopolysaccharides, proteins, and DNA, they can train and interact with the immune system to enhance both innate and adaptive immune responses when encountering pathogens in the future (Fig. 6) [35]. This mechanism aims to reduce the frequency and severity of recurrent RTIs, particularly in vulnerable populations. According to the feedback, OM-85 demonstrated a favorable safety profile, with no reported increase in the risk of adverse immune responses, supporting its suitability for long-term use in adults. Additionally, some immunological changes and improvements in lung function were observed [33].

Furthermore, bacterial lysates can be used for cancer treatment. For example, it has been demonstrated that using an engineered synchronized lysis circuit (SLC) in *Salmonella typhimurium* can control the bacterial population and facilitate the release of therapeutic agents, such as toxins that

target cells. The engineered *Salmonella* strains were tested in mouse models, improving survival in the mice [30]. Another example involves engineered *Salmonella* designed to undergo lysis after invading targeted cells. The lysis is triggered by the activation of specific genes once the bacteria are inside cancer cells. The bacterial lysis results in the release of synthesized therapeutic proteins directly into the cytoplasm of the cells. Initially, these therapeutic proteins, such as GFP, remain within the *Salmonella*-containing vacuoles (SCVs), but eventually spread throughout the cytoplasm, reaching their intended intracellular targets [36].

3.3. Bacterial vesicles and cell envelopes

In recent decades, bacterial vesicles have gained more attention as a novel therapeutic tool. Bacterial vesicles are tiny nanoparticles capable of carrying biological molecules and cellular components. They can be useful in the treatment and diagnosis of infectious diseases due to their roles in cell-to-cell communication, gene transfer, and the transport of antigens and toxins. Gram-negative and gram-positive bacteria have different cell walls and therefore produce various types of vesicles. The vesicles released from gram-negative bacteria are known as outer membrane vesicles (OMVs), produced by outer membrane budding and explosive cell lysis. Vesicles in gram-positive bacteria are referred to as bacterial membrane vesicles (BMVs). Due to the thick peptidoglycan layer, the biogenesis mechanism of BMVs remains unclear. The differences in vesicle formation and content supported the potential of bacterial vesicles in therapeutic applications, making them a promising area of research for the treatment and diagnosis of various diseases. There are some therapeutic applications of vesicles in the prevention and treatment of infectious diseases [37].

A recent study engineered OMVs from mutant *Salmonella enterica* serovar Typhimurium strains to display key *Helicobacter pylori* antigens (UreB, CagA, and VacA). These OMVs significantly increased IgG and IgA levels, and effectively stimulated Th1 and Th17 immune responses. In mouse models, co-delivery of UreB and CagA by these OMVs provided protection against *H. pylori* and induced specific T-

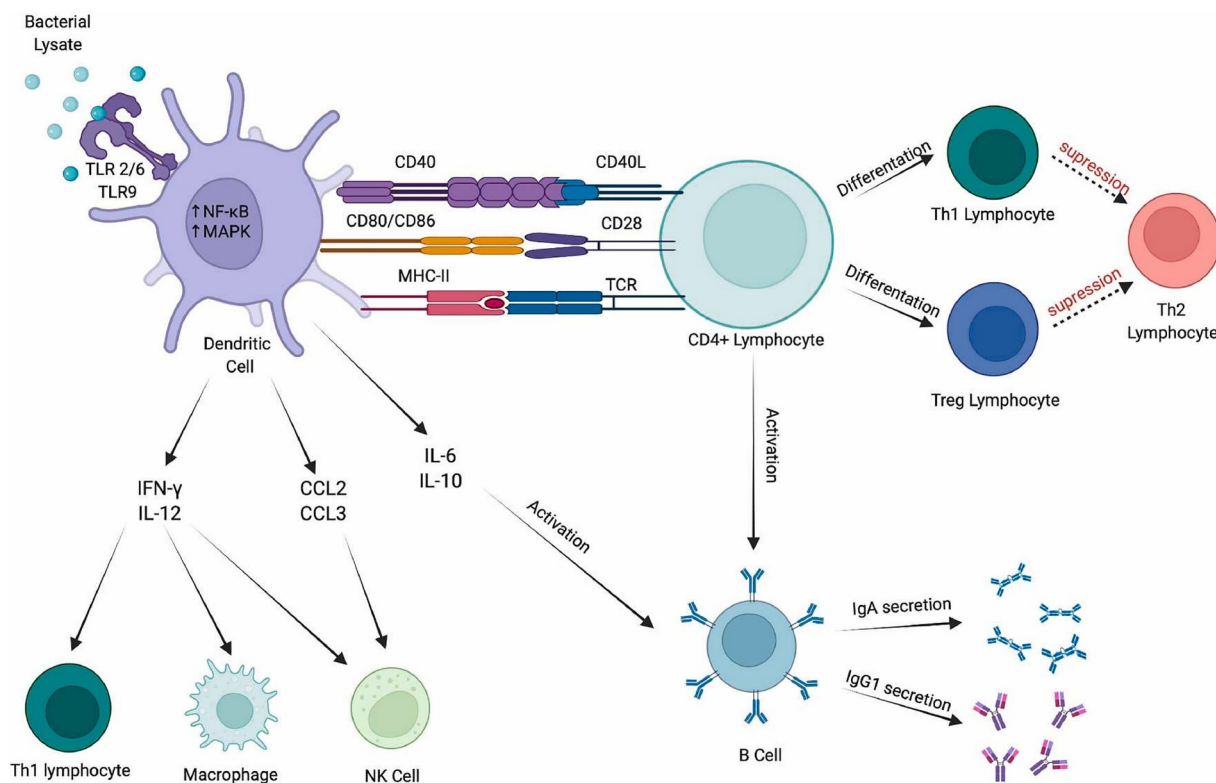


Fig. 6.

cell responses, confirming their potential as a vaccine platform for combating *H. pylori* infections [38].

In another study, researchers tested Salmonella-derived OMVs displaying *Chlamydia trachomatis* antigens (CTH522 or HtrA) as intranasal vaccines in mice. The OMV-based vaccines induced both systemic and vaginal humoral responses against the antigens. Additionally, the vaccines triggered strong antigen-specific Th1 and Th17 cellular responses, including IFN- γ production by CD8 + T cells in the spleen [39]. In a recent study, researchers evaluated the use of *Neisseria meningitidis* OMVs as an adjuvant for a SARS-CoV-2 vaccine. Mice were immunized with the receptor-binding domain (RBD) of SARS-CoV-2 combined with OMVs and aluminum hydroxide. The results demonstrated that the combination enhanced the humoral response, producing high-avidity antibodies against both SARS-CoV-2 and OMVs, and induced a strong mixed Th1/Th2 cellular immune response. Notably, a lower dose of RBD yielded a superior cellular response (Fig. 7) [40]. The study also confirmed that antibodies were effectively transferred from mothers to offspring, emphasizing OMVs' potential to boost vaccine efficacy and provide dual protection [41].

The vaccine candidate FINLAY-FR-1 includes OMVs from *N. meningitidis*. These vesicles are made up of proteins, lipids, and sugars, with components like lipopolysaccharides that help boost the immune response. In this study, 60 volunteers aged 19–59 were divided into three groups to test the safety and immune effects of two COVID-19 vaccines based on the SARS-CoV-2 RBD. One group received FINLAY-FR-1 (50 μ g RBD with OMVs), while the other groups received FINLAY-FR-1A with either 50 μ g or 25 μ g RBD, without OMVs. Each group got three doses over several weeks, and the results showed that the OMVs improved the vaccine's ability to produce strong neutralizing antibodies, making it highly effective and safe for use [42].

3.4. Bacterial spores

Some bacteria can produce spores under harsh environmental conditions. These spores are dense, circular, or oval structures that are highly resistant to physical and chemical stress. Spores can remain dormant and viable for years. Endospore-producing pathogenic microorganisms contribute to their widespread dissemination. The unique properties of spores have led to the development of stable vehicles for drug delivery and advanced biotechnological applications [43]. One study investigated the development of a novel mucosal vaccine against *Mannheimia haemolytica*, the main cause of bovine respiratory disease (BRD), using *Bacillus subtilis* spores as an adjuvant. A chimeric protein (MhCP) containing neutralizing epitopes from the *M. haemolytica* leukotoxin (NLKT) and the outer membrane protein PlpE was produced and then loaded onto *B. subtilis* spores. The results showed that the mucosal vaccine induced a strong immune response by increasing serum IgA and IgG antibody levels [44].

In a clinical trial to determine the role of *Bacillus clausii* spores (LiveSpo CLAUSY) in treating persistent diarrhea in children, researchers used higher doses (8 to 12 billion CFU/day) than usual. Results from the study in 100 children aged 3 to 24 months showed that taking this probiotic improved diarrhea 3 days faster than the control group. There were also significant decreases in some immune markers (TNF- α , IL-17, and IL-23) and decreased levels of IgA in the stool, suggesting that probiotics help reduce inflammation and support the immune system [45]. In another clinical trial, researchers evaluated the effect of administering VOWST, an

oral probiotic composed of *Firmicutes* spores, on antimicrobial resistance genes (ARGs) in patients with recurrent *Clostridium difficile* infection (rCDI). Analysis of patients' stool samples at 6 months showed that VOWST treatment significantly reduced ARG levels compared to the control group. Thus, VOWST is also effective in restoring the gut microbiome and accelerating recovery [46]. In a study to target porcine reproductive and respiratory syndrome (PRRS), researchers developed an oral vaccine using *B. subtilis* spores that provided protection against both the European (EU) and North American (NA) strains of the PRRS virus (PRRSV). The vaccine increased the production of IgG and IgA antibodies and enhanced immunity in the lungs and gastrointestinal tract by increasing IFN- γ and IL-4 [47].

In another study, *B. subtilis* spores were tested for their ability to deliver antigens to mucosal surfaces and activate immune responses in mice against H1N1 influenza virus, respiratory syncytial virus (RSV), and SARS-CoV-2. When given intranasally, the spores improved survival and reduced disease severity by increasing immune cells in the lungs and reducing harmful inflammation. The spores also helped produce memory T cells and, when used with an intramuscular vaccine, increased antibody levels (IgA and IgG) against SARS-CoV-2 [48]. Other studies have focused on tumor targeting by bacterial spores. For example, a biomimetic nanoplateform based on the spore shell of *Bacillus coagulans* was developed to enhance antitumor immunity and improve the efficacy of chemotherapy (CDT). This structure, upon entering tumor tissue, initiates chain reactions and releases hydroxyl radicals, which not only kill cancer cells but also stimulate the immune response, including increased CD8 + T lymphocyte activity [49].

In another study, researchers treated diethylnitrosamine (DEN)-induced liver cancer in mice using *Clostridium sporogenes* endospores.

The results confirmed significant suppression of tumor proliferation and long-term immunity through stimulation of IFN- γ , NF- κ B, and TNF- α [50]. A recent example is based on synthetic spore-like particles (SSHELs). SSHELs, created using SpoVM and SpoIVA proteins from *B. subtilis*, serve as innovative platforms for vaccine delivery. These particles effectively display antigens such as the *S. aureus* alpha toxin variant (HlaH35L), showing immense promise in boosting immune responses [51].

3.5. Bacterial toxins and enzymes

3.5.1. Bacterial toxins

The increasing challenge of antibiotic-resistant infections has prompted researchers to explore bacteriocins as a therapeutic approach for treating infectious diseases. Bacteriocins, peptides produced by bacteria, can selectively target and kill pathogenic strains, offering a potential alternative to traditional antibiotics [52]. In one study, researchers engineered the bacteriocin S5-PmnH to effectively combat multidrug-resistant *Pseudomonas aeruginosa*, a

This action significantly reduces the presence of *S. aureus* and inhibits the expression of its toxins that contribute to inflammation, particularly in conditions like atopic dermatitis (AD). ShA9's dual action not only combats the pathogen directly but also enhances the host's antimicrobial defenses by increasing the expression of human antimicrobial peptides, often suppressed in AD patients due to inflammatory cytokines. Importantly, while bacteriocins can be considered bacterial toxins due to their ability to kill other bacteria, ShA9's selective action allows it to maintain a healthy balance within the skin microbiome, promoting overall skin health and offering a novel therapeutic approach for managing AD and similar infectious conditions [55].

Researchers also introduced darobactin, a novel bacteriocin derived from *Photobacterium phagocytophilum*, which specifically targets Gram-negative bacteria like *E. coli* and *K. pneumoniae*. Darobactin is ribosomally synthesized and features a unique structure with two fused rings formed post-translationally. Its mechanism of action involves binding to BamA, an essential protein for the assembly of the bacterial outer membrane, which leads to the disruption of membrane integrity and ultimately results in bacterial cell death. This specificity allows darobactin to effectively combat drug-

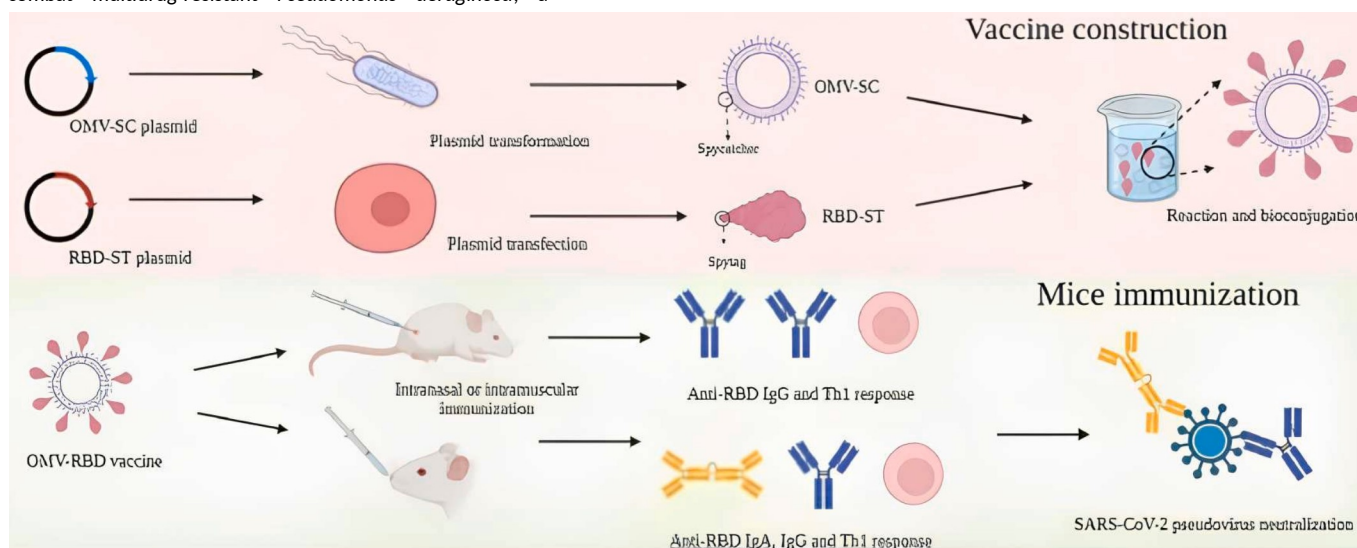


Fig. 7.

resistant strains while minimizing harm to beneficial bacteria [56].

3.5.2. Bacterial enzymes

Bacterial enzymes exhibit a wide range of functions, some of which are implicated in disease processes. Bacterial proteases can degrade host proteins, such as growth factors that help in wound healing, while simultaneously impairing immune defenses. In patients with abnormal cytological dysplasia, elevated levels of microbial enzymes, including mucinase, sialidase, and protease, have been detected, suggesting their role as cofactors in cervical cancer [57]. Research has also highlighted the beneficial aspects of bacterial enzymes. In one study, the impact of an α -amylase-producing *E. coli* strain on broiler chickens was investigated. The presence of the bacteria was found to enhance the digestibility of dry and organic matter while positively affecting the morphology of the intestinal villi. This suggests that such supplementation can improve nutrient absorption and overall chicken health [58].

common pathogen associated with severe infections. This innovative approach significantly reduced bacterial load in models of keratitis and lung infections, suggesting a new way for managing these difficult-to-treat conditions [53].

Another research focused on the probiotic *Streptococcus dentisani*, which produces bacteriocins that demonstrated effectiveness in improving oral health by displacing harmful pathogens. In a randomized, double-blind, placebo-controlled study, participants treated with a bucco-adhesive gel containing *S. dentisani* showed improved clinical and microbiological parameters associated with oral health. The application of this probiotic led to a reduction in dental plaque and gingival inflammation, alongside an increase in salivary ammonia and calcium levels, which are beneficial for maintaining oral health. The study also revealed that *S. dentisani* could effectively displace pathogenic organisms in the oral cavity, likely due to its bacteriocin production, which inhibits various cariogenic bacteria such as *Streptococcus mutans* [54].

Furthermore, studies have investigated the bacteriocin-producing strain *Staphylococcus hominis* A9 (ShA9), which produces lantibiotics that effectively kill *S. aureus* on the skin.

Bacterial enzymes also play a role in drug delivery. A glutaminase- asparaginase enzyme derived from *Achromobacter* species was found to be effective in patients with acute leukemia. Administered in varying doses, the enzyme demonstrated anti-leukemic activity by depleting levels of glutamine and asparagine [59]. In another study, collagenase derived from *Clostridium histolyticum* was applied to patients with chronic total occlusions in coronary arteries, which are blockages that hinder blood flow. Patients reported a reduction in angina symptoms, demonstrating the effectiveness of this bacterial enzyme in addressing coronary artery disease [60].

3.6. Advances in bacterial drug delivery systems

Recent advances in genetics and biotechnological techniques have led to the development of complex bacterial DDSs. A notable example is the precise manipulation of the genome of drug-delivering bacteria using CRISPR/Cas9 technology. CRISPR/Cas9, an adaptive immune effector in bacteria, consists of two gene systems within an operon. This operon encodes nucleases that form a complex with a guide RNA (sgRNA). The sgRNA recognizes the target site in the DNA, and Cas9 makes the cut (Fig. 8) [61,62]. Plasmids carrying CRISPR/Cas9 can target important virulence genes and antibiotic resistance in bacteria. Consequently, CRISPR/Cas9 has emerged as a new hope in the fight against drug-resistant bacteria and the infections they cause. Conjugative plasmids carrying CRISPR/Cas9 are often studied to combat bacterial gastrointestinal diseases, particularly those caused by *E. coli* and *Salmonella* species. Microbial transfer of these plasmids is often facilitated by bacteriophages or intestinal microbiota [63]. In one study, researchers used an *E. coli* probiotic strain carrying a conjugative CRISPR/Cas9 plasmid in a murine model to combat *Citrobacter rodentium* infection. The rate of plasmid transfer from the conjugative probiotic (COP) to other intestinal microbiota was very high, significantly inhibiting infection in mice [64]. Another study on the transformation of a plasmid carrying CRISPR/Cas into the *E. coli* EMG2 SmR strain confirmed that the system could deliver sgRNA

nucleases against carbapenem-resistant *Enterobacteriaceae*. This delivery system improved the survival rate in a *Galleria mellonella* infection model [65].

Another technology that has enhanced the efficiency of bacterial DDSs is the integration of biosensors. This allows for careful monitoring of the drug release time and the release of bacteria within the body. These systems can adapt to physiological conditions and optimize therapeutic effects. The strategy involves using engineered bacteria that respond to specific physiological conditions in the body [66]. For example, in one study, researchers designed an EcN strain equipped with eight plasmid-based sensors that were activated by molecules in the human body. The activators included acetoacetate, lactate, caffeine, taurocholic acid, deoxycholic acid, lithocholic acid, oxygen, and pH. EcN was colonized in *Caenorhabditis elegans*, and the nematode was then placed on a medium containing a biosensor-activating agent that mimicked conditions similar to the human gut. The expression of the plasmid genes equipped with the sensor was observed in the presence of the activator, confirming their functional control [67].

Challenges also exist with CRISPR technologies. Although scientists have developed arrayed CRISPR libraries using a method called ALPA (automated liquid-phase assembly) to enhance gene manipulation, constructing these libraries is labor-intensive and often yields inconsistent results due to reliance on a single promoter for sgRNA expression. Outdated reference genomes can further hinder sgRNA design in genetically diverse patient cells, highlighting the need for improved library design and cloning methods [68].

4. Administration routes of bacterial drug delivery systems

Bacterial DDSs are advanced therapeutic methods that use engineered bacteria to deliver and release medications directly at infection or tumor sites. These systems use different methods depending on therapeutic goals. Intratumoral and intravenous (IV) routes are effective for cancer treatments, oral methods support gut health, nebulization treats respiratory diseases, and topical applications suitable for skin conditions. Fecal microbiota transplantation (FMT) is

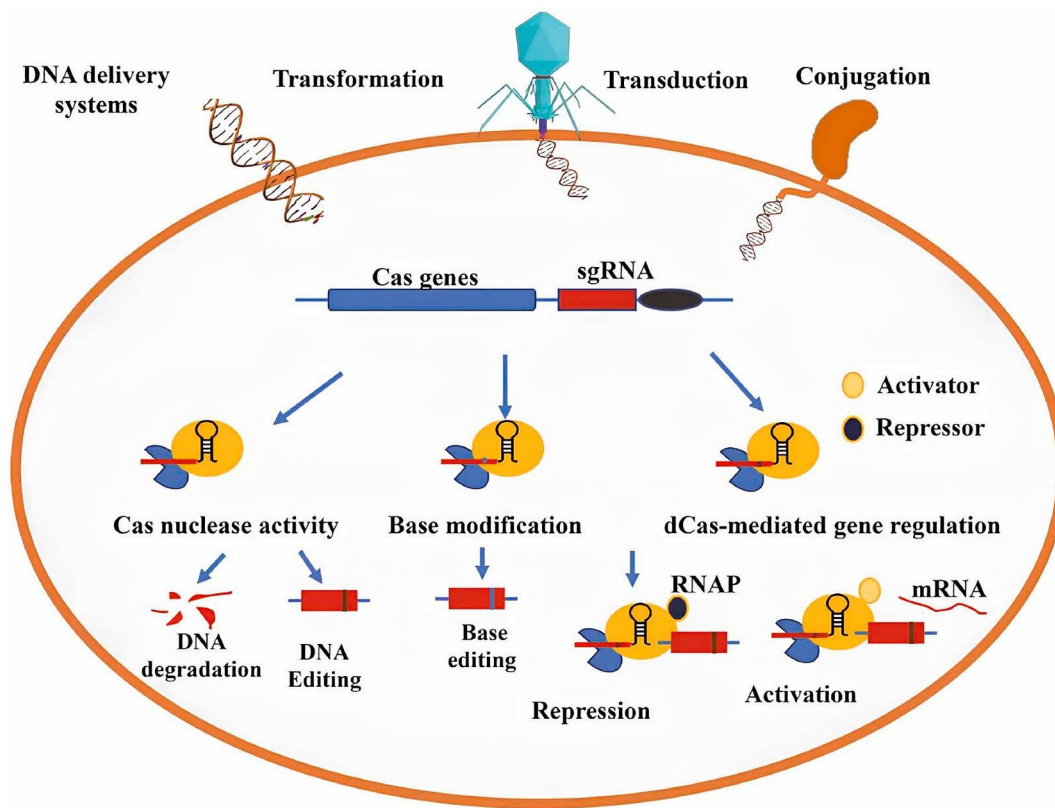


Fig. 8.

essential for restoring gut microbiota. These diverse strategies highlight the potential of bacterial therapies in medicine. The intravenous injection method allows the bacteria to circulate throughout the body and reach various tumor sites via the bloodstream. The intra-tumoral injection approach maximizes bacterial concentration at the target site, enhancing therapeutic effectiveness while minimizing systemic exposure. For example, injecting SYN1891 derived from EcN directly into tumors activates immune responses via the STING pathway, and reduces systemic exposure [69].

Nebulized bacterial lysates are inhaled directly into the lungs for localized respiratory treatments, such as COPD. This method minimizes systemic exposure and enhances treatment effectiveness [70]. Another method is topical administration. For example, heat-killed *Lactobacillus plantarum* in creams improves skin hydration and microbiome balance. This approach has shown promise for treating conditions such as eczema and acne. Another example is *L. plantarum* GMNL6, which has been shown to improve skin conditions through topical applications, supporting its potential use as a cosmetic ingredient [71].

In oral delivery, although challenging due to the acidic conditions of the gastrointestinal tract, advancements in bacterial encapsulation show potential for effective oral administration. Combination therapies of oncolytic bacteria can also be used with conventional chemotherapy or immunotherapy treatments to boost overall efficacy. Oral delivery is widely used for probiotics like *B. clausii* and *Lactobacillus acidophilus*. These bacteria survive digestion to restore gut health. For example, *Lactobacillus paracasei* has been used via oral gavage to deliver ACE2 proteins to the colon, effectively targeting hypertension [72]. Another example is *A. muciniphila*, which, by modulating gut microbiota, exhibited therapeutic stability in patients with familial Mediterranean fever [73]. *L. paracasei* also effectively lowered blood pressure in hypertensive rat models, with significant gender-specific effects in females, suggesting a role in personalized medicine [74]. Furthermore, *L. plantarum* demonstrated capabilities in improving collagen synthesis and reducing melanin production, indicating applications in skin care [71].

The final method is FMT, which restores gut microbiota by transferring healthy bacteria from donors to recipients using capsules, colonoscopy, or nasogastric tubes. It is particularly effective for treating recurrent *C. difficile* infections. For example, using FMT to increase the *Clostridium butyricum* population. It plays a role in restoring gut microbiota following gastrectomy and improving immune modulation by enhancing SCFAs production. The bacterium also reduced inflammation markers like IL-1 β and TNF- α in post-surgical gastric cancer patients, enhancing overall gut health [75].

5. Applications of bacterial drug delivery systems

5.1. Infectious disease

Infection is a significant global public health challenge, affecting millions each year. New and effective methods for delivering medicine to infection sites resistant to conventional treatments have been developed. In a recent study, researchers investigated *L. lactis* as a delivery system for the antimicrobial peptide Clf chimera, targeting foodborne pathogens like *P. aeruginosa* and *S. aureus*. The engineered strain effectively prevented biofilm formation and showed potential for infection management [76].

Another study evaluated the use of *L. lactis* N29000 as a delivery mechanism for bacteriophage lysins (Endo 88 and

VAH 88) and virion-associated peptidoglycan hydrolase (VAPGH) to combat multi-drug-resistant *S. aureus* (MRSA). The results showed that recombinant *L. lactis* successfully inhibited the growth of *S. aureus* PS88, achieving a 3.5 log reduction in bacterial counts. Additionally, Endo 88 proved to be more effective than VAH 88 in lysing MRSA strains [77].

Another study evaluated the use of DDS with *Streptomyces coelicolor* to address infections caused by multi-drug-resistant (MDR) bacteria. This approach involves co-culturing the bacterium with the fungus *Aspergillus niger*, fostering interactions that lead to the creation of new compounds. The novel antimicrobial compounds include (E)-2-(3-hydroxyprop-1-en-1-yl)-phenol and (2E,4E)-3-(2-carboxy-1-hydroxyethyl)-2,4-hexadienedioic acid. The bacteria can then be used as DDSs containing antimicrobial agents to deal with plant pathogens [78].

5.2. Gastrointestinal Disorders

Over the past decades, gastrointestinal diseases, especially colon-associated diseases, have significantly increased in the West due to conditions such as Crohn's disease, irritable bowel syndrome, colorectal cancer, and ulcerative colitis. Researchers use various methods for treating gastrointestinal diseases, including designed prophylactic drugs and different DDSs. Although these methods have positive effects on the treatment process, some life-threatening and infectious illnesses cannot be treated or managed by these drugs. These diseases cause issues during treatment, such as adverse responses to medicines. Another important point is the ability of drugs to cope with challenging conditions like a large pH gradient. To mitigate the consequences of traditional therapeutic methods, specifically targeted DDSs with fewer side effects and better functionality are used in both diagnosing and delivering therapeutic agents in a controlled manner. EcN is a strain of bacteria with antibacterial and anti-inflammatory properties, showing potential in treating various diseases, including gastrointestinal conditions. EcN is marketed as the active component in the drug Mutaflor® for the treatment of IBD [79].

5.3. Tumors

Despite extensive research, cancer remains a major threat to human health due to its high fatality rate. According to a WHO report, there were 9.6 million deaths, or roughly 1 in 6 deaths, in 2018. Various treatments for cancer, such as chemotherapy and radiotherapy, are used, but these methods have numerous side effects, primarily due to the non-targeted distribution of drugs. Advancements in tumor-targeted delivery systems are crucial for developing anti-cancer DDSs. Widely studied DDSs demonstrate some shortcomings, such as the low cancer cell targeting efficiency of nanoparticles [80]. Studies have shown that approximately 0.7 % of nanoparticles are delivered to a solid tumor. Hence, the development of effective DDSs is essential to address these issues. Recently, extensive investigations into bacterial-based biocarriers have been conducted due to their proper tumor targeting, immunomodulatory, and oncolytic properties [15].

Bacterial-based drug carriers are considered a new type of anticancer treatment that can accumulate more easily in tumor tissue than in healthy tissues. The tumor-targeting characteristic is related to the suitable tumor microenvironment (TME) and the bacteria's features [81]. Another approach is the immunomodulatory effects of bacterial carriers. Enhancing immune responses, such as

activating inflammasome pathways and reducing immune escape, are two main ways of activating the immune system against tumors by bacterial-based drug carriers. These carriers can effectively stimulate the host immune system [82].

Finally, the use of oncolytic bacterial carriers involves bacteria accumulating in tumors to activate the immune system by killing cancer cells. This process causes anticancer effects that enhance the role of drugs in treatment. Different pathways have been observed through which these oncolytic activities can be achieved, such as reducing tumor drug resistance and inhibiting tumor metastasis [60]. Notable examples include *Clostridium* and *Salmonella* species, which are important types of bacteria from different groups. They have demonstrated efficient roles in tumor targeting, as proven by various experiments. *Clostridium* is an obligate anaerobic bacterium used in Malmgren's experiment, showing that this bacterium specifically accumulates in tumors, not in healthy tissues, because tumors have hypoxic regions [83]. *Salmonella*, a facultative anaerobic bacterium, targets tumors using different mechanisms, such as preferential growth in tumor environments and attraction toward tumors through inflammatory responses [84].

A study highlighted the role of flagella in *Salmonella*-mediated cancer therapies in activating the host immune response. The antitumor pathway of flagellin is shown in Fig. 9, which, by stimulating signaling through TLR5, ultimately leads to the activation of T lymphocytes against cancer cells (Fig. 9) [85]. By stimulating the Flagellin/TLR5/NF- κ B pathway, flagella enhance immune activation, which is essential for the anti-tumor effects of *Salmonella* [86]. Another example is the use of *L. monocytogenes* in a clinical trial for patients with advanced melanoma. The bacterial treatment effectively triggered an immune response against the tumor, leading to substantial tumor shrinkage in several patients [87]. In another study, engineered *S. typhimurium* (designated as SAM-FC) was utilized to deliver dual therapeutic agents, including cytolysin A (ClyA), a cytolytic anti-cancer compound, and *Vibrio vulnificus* flagellin B (FlaB), to enhance anti-tumor innate immunity [88].

Another study examined the potential of the bacterial effector protein HopQ from *Pseudomonas syringae* (Pto) as a therapeutic agent against melanoma metastasis. The type III secretion system allows bacteria to inject HopQ directly into the cytoplasm of melanoma cells. This interaction leads to the degradation of vimentin, resulting in decreased motility of melanoma cells and a reduction in metastasis [89]. Another study indicated the function of heparin, an anticoagulant medication, produced by *E. coli* K5 from specific N-trifluoroacetyl-glucosamine. They enhanced the production method of heparin, which needs to be purified and transformed into an appropriate pharmaceutical form [90]. Genetically engineered *E. coli* producing gas vesicles enhance precision in focused ultrasound ablation surgery (FUAS) by serving as cavitation nuclei within tumors [91]. Another example is CRS-207, a live-attenuated *Listeria monocytogenes* engineered to express mesothelin (MSLN), designed to trigger targeted immune responses against cancer [92].

6. Challenges, safety, and ethical considerations

Before starting the use of bacterial DDSs in patients, several factors must be considered. First, informed consent is essential; patients must fully understand the risks, benefits, and uncertainties of the trial. Additionally, the research team must ensure that patients are not pressured into participating, particularly in urgent situations. For example, a Phase II trial of Axalimogene Filolisin (ADXS-HPV) emphasized informed

consent, rigorous safety protocols, and conflict of interest transparency, demonstrating key ethical practices to protect patient welfare in clinical research. Similarly, in bacterial research, studies often use controlled bioreactors to manage risks associated with resistant strains, ensuring environmental safety while advancing the understanding of resistance management strategies [93].

Despite significant advances and many advantages in bacterial DDSs, there are still numerous challenges in applying these tools in clinical use. The most important concern regarding bacterial DDSs is the use of live bacteria or bacterial components in the human body. The possibility of infection, induction of destructive immune responses, continuation of inflammatory pathways, spreading through the body, and changes in natural microbiota are some of the most critical challenges associated with bacterial DDSs. These

the stability of the secreted protein. By optimizing fermentation and using microencapsulation, they preserved the protein's bioactivity and facilitated effective oral delivery, showcasing a promising strategy for treating liver fibrosis [96].

The use of genetic engineering technologies can also help personalize medicine. Bacterial DDSs designed based on the genetic and metabolic characteristics of patients can increase treatment effectiveness and reduce side effects. Despite the difficulty and high cost of producing these strains, obtaining a

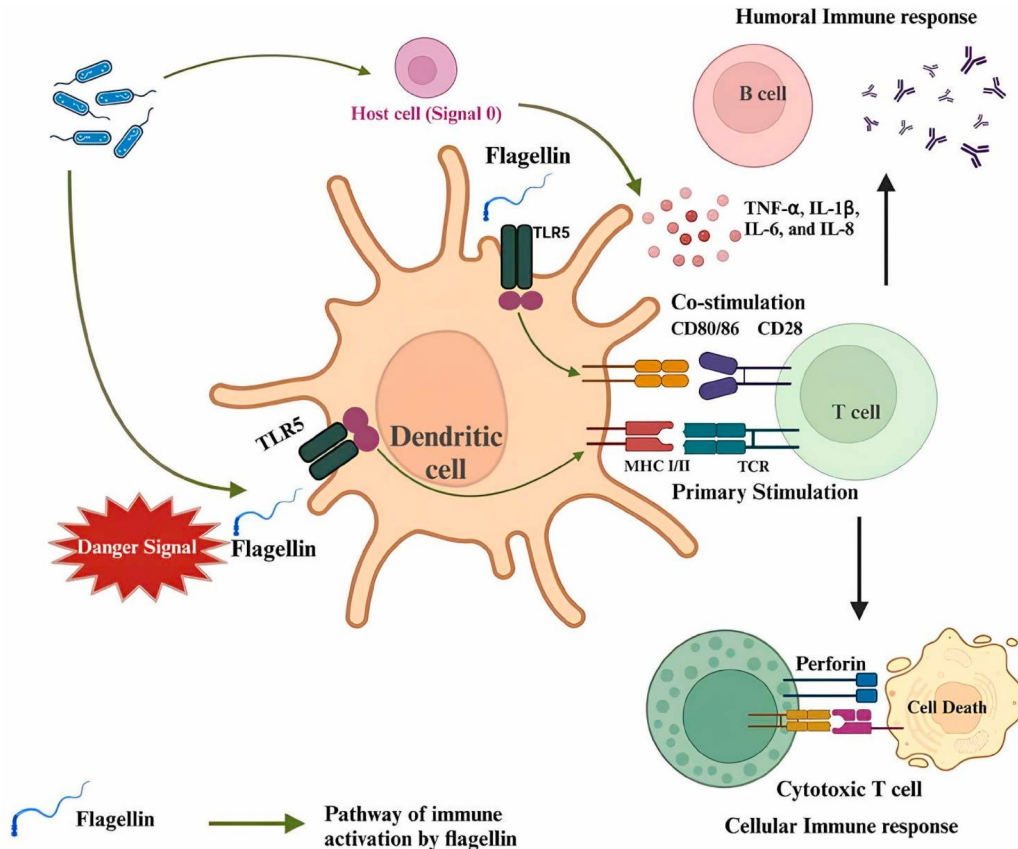


Fig. 9.

complications are especially important in individuals with underlying diseases, as they can expose them to life-threatening conditions such as sepsis. Additionally, the change in microbiota due to the use of bacterial DDSs has received less attention. The microbiota, especially in the human gut, plays a key role in modulating immune responses and the biosynthesis of essential substances such as folic acid, short-chain fatty acids (SCFAs), and vitamins. Therefore, the occurrence of dysbiosis disrupts all these pathways [94].

A solution to creating a safe bacterial DDS is to use engineered strains. Engineered strains not only carry drugs but also have repressed virulence genes and reduced expression of pathogen-associated molecular patterns (PAMPs). Consequently, the immunogenic properties of bacterial DDSs are controlled, and the intensity of destructive immune responses is reduced. Additionally, the possibility of targeted drug delivery and controlled activity of the carrier bacteria is provided [83]. For example, a study emphasized the importance of understanding host immune responses in *Salmonella*-mediated therapies, as different bacterial strains can produce varying cytokine profiles that affect immune response and tumor suppression [95]. Scientists can also employ techniques to overcome these challenges. For example, in developing a *L. lactis* strain that expresses TGFβR2 to alleviate liver fibrosis in mice, researchers faced issues with

license for genetically modified organisms (GMOs) is challenging [97]. Another approach is to use non-pathogenic strains such as probiotic bacteria. This method not only leverages the benefits of probiotic consumption but also achieves therapeutic goals using bacterial DDSs. Furthermore, the risk of dysbiosis and harmful immune system reactions is reduced [98]. Therefore, the application of bacterial DDSs from the laboratory bench to the bedside requires interdisciplinary collaboration between microbiologists, biotechnologists, and clinicians to manage the multifaceted challenges associated with bacterial drug delivery systems.

7. Conclusion

This review has provided an overview of the current state of knowledge regarding bacterial DDSs against infectious diseases. Bacterial DDSs represent cutting-edge therapeutic approaches that leverage the unique properties of engineered bacteria to deliver and release medications directly at infection or tumor sites. Despite the significant biotechnological development and refinement of bacterial DDSs by means of genetic engineering, many challenges must be overcome for clinical translation of this technology. The most critical issues to be taken into consideration involve the effects that DDSs could have on the host's immune system

and microbiome. Infection, induction of destructive immune responses, perpetuation of inflammatory pathways, and alteration of natural microbiota-all of these might carry a great risk, especially for patients with underlying conditions. Integration of technologies in genetic engineering will further make medicine personalized, enhancing its effectiveness and minimizing side effects.

The further study should enhance our knowledge in terms of deeper insight into a very precise in-vivo role of DDS bacterial carrier and include immunomodulatory activities, oncolytic activities using bacterial carriers, development of synthetic spore-like nanomaterial and other new interesting delivery systems. Complex problems concerning all aspects of BDS are best solved in collaborations among microbiologists, biotechnologists, and clinicians in close contact, discussing and contributing views that open future clinical perspectives in the field. By advancing our knowledge and overcoming these challenges, bacterial DDSs hold great promise for improving the treatment of infectious diseases, cancer, and other medical conditions, ultimately enhancing patient outcomes and quality of life.

CRedit authorship contribution statement

Parastoo Tabibzadeh Tehrani: Writing – original draft, Software. **Mina Nazari:** Writing – original draft. **Pedram Rastgoo:** Writing – review & editing. **Nilofar Seyed Bolouri:** Writing – review & editing. **Reyhaneh Heydari Karsaf:** Writing – original draft. **Abtin Hadiani:** Writing – original draft, Software. **Zeinab Mohsenipour:** Writing – review & editing, Writing – original draft, Supervision, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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None.

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